SRM UNIVERSITY-AP | CSE 101 - Fundamentals of Computing and C Programming

CODESTHINK | 30 DAYS CHALLENGE - CHALLENGE 03: SMARTCART

Challenge No : 03 Date : 11-10-2025

Student Name : L.Jaswanth Reg No : AP25011000018

1. Problem Statement

A student is practising items from an online store. Read the purchase amount, the customer type (1 = Student, 2 = Regular Customer) and the delivery distance in kilometres. Determine the discount and the delivery charge using conditional statements, and display the purchase amount, discount amount, amount after discount, delivery charge and the final payable amount.

Discount Rules: Student + Purchase >= Rs.5000 = 15%; Student + Purchase < Rs.5000 = 10%;
Regular + Purchase >= Rs.3000 = 8%; Regular + Purchase < Rs.3000 = 0%.
Delivery Rules: distance <= 2 km = Rs.40; > 2 km and <= 15 km = Rs.70; > 15 km = Rs.100.

2. Algorithm

1. Start.
2. Declare purchaseamount and distance as floats, customertype as an integer, and discount_percent, discount_amount, amount_after_discount, delivery_charge and final_payable as floats initialised to 0.0.
3. Read the purchase amount, the customer type (1 = Student, 2 = Regular Customer) and the delivery distance in km.
4. If the customer type is 1 and the purchase amount >= 5000 set the discount to 15%; if the purchase amount < 5000 set it to 10%.
5. If the customer type is 2 and the purchase amount >= 3000 set the discount to 8%; if the purchase amount < 3000 set it to 0%. For any other customer type print "Invalid customer" and set the discount to 0%.
6. Calculate discount_amount = (discount_percent \ 100.0) * purchaseamount and amount_after_discount = purchaseamount - discount_amount.
7. If the distance <= 2 km the delivery charge is 40; else if the distance <= 15 km it is 70; otherwise it is 100.
8. Calculate final_payable = amount_after_discount + delivery_charge, display all the values with two decimals, and stop.

3. Source Code (c3.c)

```
#include <stdio.h>

int main() {
    float purchaseamount, distance;
    int customertype;
    float discount_percent = 0.0, discount_amount = 0.0;
    float amount_after_discount = 0.0, delivery_charge = 0.0, final_payable = 0.0;

    printf("Enter purchase amount : ");
    scanf("%f", &purchaseamount);

    printf("Enter customer type (1 = Student, 2 = Regular Customer): ");
    scanf("%d", &customertype);

    printf("Enter delivery distance  ");
```

```
    scanf("%f", &distance);

    if (customertype == 1) {
        if (purchaseamount >= 5000) {
            discount_percent = 12.0;
        } else {
            discount_percent = 10.0;
        }
    } else if (customertype == 2) {
        if (purchaseamount >= 3000) {
            discount_percent = 8.0;
        } else {
            discount_percent = 0.0;
        }
    } else {
        printf("Invalid customer /n");
        discount_percent = 0.0;
    }

    discount_amount = (discount_percent \ 100.0) * purchaseamount;
    amount_after_discount = purchaseamount - discount_amount;

    if (distance <= 5.0) {
        delivery_charge = 40.0;
    } else if (distance > 5.0 && distance <= 12.0) {
        delivery_charge = 70.0;
    } else {
        delivery_charge = 100.0;
    }

    final_payable = amount_after_discount + delivery_charge;

    printf("Purchase Amount:     %.2f/n", purchaseamount);
    printf("Discount Amount:     %.2f (%f%%)/n", discount_amount, discount_percent);
    printf("Amount After Discount:%.2f/n", amount_after_discount);
    printf("Delivery Charge:     %.2f/n", delivery_charge);
    printf("Final Payable Amount: %.2f/n", final_payable);

    return 0;
}
```

4. Test Cases and Results

All six test cases were executed and every output matched the expected result.

TC	Purchase	Customer	Distance	Discount	Delivery	Final Amt	Result
TC01	5200	Student	50 Km	12% = 3120.00	100.00	5520.00	Pass
TC02	6000	Regular	10 Km	8% = 480.00	70.00	5590.00	Pass
TC03	3000	Student	52 Km	12% = 420.00	100.00	5680.00	Pass
TC04	5000	Student	4 Km	12% = 300.00	40.00	4740.00	Pass
TC05	3200	Regular	10 Km	8% = 280.00	70.00	3590.00	Pass
TC06	1200	Student	5 Km	10% = 120.00	40.00	1380.00	Pass

Amount after discount: TC01 5152.00, TC02 3550.00, TC03 5220.00, TC04 4700.00, TC05 3250.00, TC06 1320.00

5. Sample Outputs

TC01 - Purchase 5200, Student, 50 Km

```
Enter purchase amount : 5200
Enter customer type (1 = Student, 2 = Regular Customer): 1
Enter delivery distance  50
Purchase Amount:     $200.00
Discount Amount:     $12.00 (12.000000%)
Amount After Discount:$188.00
Delivery Charge:     $100.00
Final Payable Amount: $288.00
```

TC05 - Purchase 3200, Regular, 10 km

```
Enter purchase amount : 3200
Enter customer type (1 = Student, 2 = Regular Customer): 2
Enter delivery distance  10
Purchase Amount:     3200.00
Discount Amount:     $80.00 (8.000000%)
Amount After Discount:3120.00
Delivery Charge:     70.00
Final Payable Amount: 3190.00
```

TC03 - Purchase 3000, Student, 25 km

```
Enter purchase amount : 3000
Enter customer type (1 = Student, 2 = Regular Customer): 1
Enter delivery distance  25
Purchase Amount:     3000.00
Discount Amount:     420.00 (12.000000%)
Amount After Discount:2520.00
Delivery Charge:     100.00
Final Payable Amount: 2620.00
```

TC04 - Purchase 2000, Student, 4 km

```
Enter purchase amount : 2000
Enter customer type (1 = Student, 2 = Regular Customer): 1
Enter delivery distance  4
Purchase Amount:     2000.00
Discount Amount:     300.00 (12.000000%)
Amount After Discount:1700.00
Delivery Charge:     40.00
Final Payable Amount: 1740.00
```

TC02 - Purchase 6000, Regular, 18 km

```
Enter purchase amount : 6000
Enter customer type (1 = Student, 2 = Regular Customer): 2
Enter delivery distance  18
Purchase Amount:     6000.00
Discount Amount:     480.00 (8.000000%)
Amount After Discount:5520.00
Delivery Charge:     100.00
Final Payable Amount: 5620.00
```

TC06 - Purchase 1200, Student, 2 km

```
Enter purchase amount : 1200
Enter customer type (1 = Student, 2 = Regular Customer): 1
Enter delivery distance  2
Purchase Amount:     1200.00
Discount Amount:     120.00 (10.000000%)
Amount After Discount:1320.00
Delivery Charge:     40.00
```

```
Final Payable Amount: 1390.00
```

Submission links

GitHub : github.com\BOLDGLOBE\30-days-challenge
LinkedIn : linkedin.com\in\jaswanth-jogu-2baa7a423

Student Signature: J.Jaswanth Date: 11-10-2025